

NATIONAL OCCUPATIONAL STANDARD FOR CIVIL ENGINEER



NOS.CE.01
FIRST EDITION

APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 20th December, 2023.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Construction NOSDT, upon which the following organisations were represented:

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2. Engineering Institution of Zambia (EIZ)
3. Ministry of Local Government and Rural Development (MLGRD)
4. Road Development Agency (RDA)
5. Surveyors Institute of Zambia (SIZ)
6. University of Zambia (UNZA)
7. Zambia Institute of Architects (ZIA)
8. Association of Building and Civil Engineering Contractors (ABCEC)
9. Water Resources Management Authority (WARMA)
10. Bari Zambia Limited (BZL)
11. Zulu Barrow Construction (ZBC)
12. Ng'andu Consulting (NC)
13. Zambia Qualifications Authority (ZAQA) – Secretariat

ACKNOWLEDGEMENT

The Zambia Qualifications Authority would like to acknowledge the invaluable support of the following stakeholders that participated in the development of this NOS:

1. Arch. Mofya Kumisuku (Zambia Institute of Architects)
2. Arch. Peter Simphila Chiuto (Zambia Institute of Architects)
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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Construction National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Construction sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

A Civil Engineer plays a critical role as he/she harnesses natural resource for the universal benefit of mankind in both the public and commercial sectors, as he/she conceptualises, designs, develops, oversees, manages, and maintains infrastructure projects and systems, such as highways, buildings, airports, tunnels, dams, bridges, and water supply and sewage treatment systems. This National Occupational Standard highlights the core knowledge, skills, competences and personal attributes that Civil Engineers must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

CE	Civil Engineer
CS	Core Skills
EIZ	Engineering Institution of Zambia
K	Knowledge and Understanding
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Context
OSHE	Occupational Safety, Health and Environment
PC	Performance Criteria
PPP	Public Private Partnership
PS	Professional Skills
RK	Regulatory Context
RPL	Recognition of Prior Learning
S	Skills
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZPPA	Zambia Public Procurement Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Disability: is the physical or mental impairment that substantially limits one or more major life activities.

Function: is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

Elements: set out competences the incumbent should possess to carry out the day-to-day activities.

1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS, and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.CE.01
Occupation	Civil Engineering
Job Title	Civil Engineer
Job Description	A Civil Engineer plans, designs and oversees construction and maintenance of building structures and infrastructure, such as roads, railways, airports, bridges, harbours, dams, irrigation projects, power plants, and water and sewerage systems, etc.
Job Purpose	Civil Engineers work in a variety of industries. Civil Engineers play a fundamental role in making the world more liveable by building bridges, road, airports, homes, hospitals, parks, tunnels, stadiums, water and wastewater treatment plants, power generation stations, flood control systems, irrigation schemes, etc. This is achieved by harnessing nature for the universal benefit of mankind.
ZQF Level	Level 7
Sector	Construction
Sub sectors	Building Works, Roads and Highways, Bridges, Railways, Airports, Ports, Renewable Energy, Infrastructure, Water and Sanitation, Dams, Power Stations, Tunnels, Irrigation, etc.
Other Economic Sector(s) in which the Occupation is Practiced	Education, Mining, Transport, Telecommunication, Health, water, and Energy Sector, etc.
Other Similar Jobs that can be Performed in the Occupation	Project Engineer, Research Engineer, Maintenance Engineer/Manager, Tutor/Lecturer/Trainer or Training Manager, Sales Engineer/Manager, Consultant, Construction Dispute Resolution Board, Quantity Surveying, Town and Country Planning, Environmentalist, Archaeologists, Mining Sector, etc.
Minimum Educational Job Entry Qualification(s)	Bachelor's degree in Civil Engineering, or equivalent
Practicing License Requirements (if any)	Membership with the Engineering Institution of Zambia (EIZ) and Practicing License from the Engineering Registration Board (EngRB).
Training/RPL	1. Previous Diploma in Civil Engineering with a minimum work experience of 10 years

	2. Awareness of the Industry Standards and Rules and Regulations and their application 3. Use of ICTs (Internet, Computer packages, email, Computer Software and Hardware necessary for the job, etc.). 4. Quality Enhancement Methods. 5. General industry practice (site and office work)
Minimum Job Entry Age	21 years
Prior Experience (Suggested)	Minimum of 6 months internship in construction, consultancy, or relevant government agencies
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This NOS highlights the core competences, skills, knowledge and understanding, and personal attributes that Civil Engineers are required to possess to be successful in their jobs.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess the following attributes:

- Creativity
- Critical Thinking
- Technical Competence
- Problem-solving skills
- Project Management skills
- Effective leadership
- Communication skills
- Interpersonal Skills
- Integrity
- Project Investment and Finance Awareness
- Economic Awareness
- Green Industrial Revolution Awareness
- Entrepreneurship
- Occupational Safety and Environmental awareness, etc.

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 5 units representing the tasks that a jobholder should undertake in his/her day-to-day work. The unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This Unit covers the skills and knowledge required by a Civil Engineer to undertake conceptualising, planning, and designing of civil engineering works]

Unit No.	01
Unit Title	Conceptualise/Plan and Design
Description	This Unit describes the skills and knowledge required to by a Civil Engineer to conceptualise, plan and design civil engineering works.
Scope	This Unit covers the following: • Conceptualising the Design as Given by the Client/Owner's Technical Brief/Terms of Reference. • Assessing Site Conditions. • Key Stakeholder Management. • Planning and Design.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Conceptualising the Design as Given by the Client/Owner's Technical Brief/Terms of Reference	To be competent, the individual must be able to: PC1. Understand the client/owner's project requirements, and conceptualise as required. PC2. Perform preliminary analysis of the project requirements. PC3. Conduct document review.
Assessing Site Conditions	To be competent, the individual must be able to: PC4. Perform a site reconnaissance survey. PC5. Undertake topographical/geotechnical and materials investigations. PC6. Conduct laboratory material testing. PC7. Carry out hydrological studies.
Key Stakeholder Management	To be competent, the individual must be able to: PC8. Engage key stakeholders of the project (government officials, local leaders, and other interest groups).
Planning and Design	To be competent, the individual must be able to: PC9. Undertake pre-feasibility/feasibility study. PC10. Conduct document review of similar aspects. PC11. Generate designs (structural, highways, water and sanitation, etc.). PC12. Generate technical drawings, details and associated schedules. PC13. Take-off quantities and generate the Bill of Quantities. PC14. Formulate cost estimates. PC15. Demonstrate appreciation of basic economics with respect to construction and other associated sectors. PC16. Demonstrate comprehension of project investment and finance. PC17. Prepare technical specifications. PC18. Compile tender documentation.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: - OK1. Relevant standards, procedures, and policies of the organisation. - OK2. Context of the organisation as determined by external factors, such as legal, financial, social, regulatory, and cultural. - OK3. Internal factors, organisational structure, governance, and resource capabilities. - OK4. Roles and responsibilities. - OK5. Risk management. - OK6. Occupational safety, health, quality, and environmental management. - OK7. Performance evaluation. - OK8. Change order management.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Local construction codes and mandatory standards. - TK2. Engineering design principles. - TK3. Relevant design codes and standards. - TK4. Draughting and detailing of technical drawings. - TK5. Computer literate in Computer Aided Design (CAD) software - TK6. Civil engineering standard form contract types. - TK7. Perform and/or supervise materials testing both in-situ and laboratory. - TK8. Relevant technical specifications for materials, performance, and conformity. - TK9. Off-take of quantities from drawings. - TK10. Basic risk management. - TK11. Usage of drones for engineering applications. - TK12. Technical knowledge of various construction materials. - TK13. Green industrial revolution. - TK14. Basic negotiations strategies. - TK15. Human resource management. - TK16. Collaboration with other disciplines such as mechanical, electrical/electronics, geomatic engineering, Quantity Surveyors, Architects, Environmentalists, Archaeologists, etc.
C. Regulatory Context (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and have understanding of regulatory compliance aspects of the industry pertaining to latest laws, regulations, guidelines, and specifications, but not limited to: - RK1. Engineering Institution of Zambia Act, 2010. - RK2. Mines and Minerals Development Act, 2015. - RK3. Environmental Management Act, 2011. - RK4. Occupational Health and Safety Act, 2010. - RK5. Factories Act Cap 441.

	RK6. Workers Compensation Act No. 10 of 1999. RK7. Employment Act Chapter 268, and associated labour legislature. RK8. Public Roads Acts, No. 12 of 2012. RK9. Tolls Act 2011. RK10. Public Procurement Act 2020, etc.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Produce technical reports. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers/others. CS3. Develop/review standard particular specifications. CS4. Develop/review particular contract conditions. CS5. Communicate effectively through writing as appropriate for the benefit of the audience. CS6. Use conversational communication methods such as e-mail, WhatsApp, etc. CS7. Demonstrate good command of the English language.
	Reading Skills
	The individual on the job must be able to: CS8. Read English proficiently, and/or be able to give instructions in the local language used in the office or at the site. CS9. Read and interpret sketches, drawings, schedules, and technical instructions provided for the required work. CS10. Research, read, and interpret technical data from manuals, books, and other relevant literature. CS11. Read and comprehend written information or communication.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS12. Demonstrate active listening and interpret communication/instructions from the supervisor and other co-workers. CS13. Convey information clearly and concisely to co-workers and others. CS14. Illustrate ideas using presentation skills and applications such as PowerPoint, etc. CS15. Manage technical meetings and discussions. CS16. Provide feedback on technical works and reports.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Adhere to the organisation's decision-making policies. PS2. Make independent and sound decisions based on engineering judgement, and considering public safety and interest of other stakeholders.

	PS3. Make prompt decisions on designs, plans etc. when required.
	Plan and Organise
	The individual on the job should be able to:
	PS4. Plan, organise, lead, and control organisation activities in line with the applicable deadlines.
	PS5. Schedule tasks and complete them within allocated time and budget.
	PS6. Work to achieve set goals.
	PS7. Evaluate the Strengths, Weaknesses, Opportunities and Threats (SWOT) of the task, assignment, and/or organisation.
	Customer Centricity
	The individual on the job should be able to:
	PS8. Manage customer relationships with intent to satisfying their project requirements.
	Problem-Solving
	The individual on the job should be able to:
	PS9. Process, gather, analyse, and communicate information to identify and troubleshoot solutions.
	PS10. Demonstrate ability to identify problems, brainstorm, and analyse possible results with the view of implementing the optimal solution.
	PS11. Consult widely and identify possible remedies.
	PS12. Escalate when required in line with the organisation's procedures and protocols.
	Analytical Thinking
	The individual on the job should be able to:
	PS13. Apply domain knowledge, observations, and data to perform tasks related to the assignment.
	PS14. Apply a methodical step-by-step approach to thinking and breaking down complex problems into smaller and manageable components.
	PS15. Demonstrate creativity, interpersonal and organisational skills.
	PS16. Employ data analysis, logical thinking, research, and communication skills.
	Critical Thinking
	The individual on the job should be able to:
	PS17. Demonstrate ability to observe and predict opportunities, threats, and solutions.
	PS18. Collect, understand, interpret data and other information.
	PS19. Demonstrate ability to draw inferences based on relevant data, personal knowledge, and experience.
	PS20. Communicate resulting information with others verbally, non-verbally, and in writing.

UNIT 2 [This Unit covers the skills and knowledge required by a Civil Engineer to undertake procurement of engineering services, goods, and works]

Unit No.	02
Unit Title	Procurement of Engineering Services, Goods, and Works
Description	This Unit describes the skills and knowledge required by a Civil Engineer for the procurement of engineering services, goods, and works in civil engineering.
Scope	This Unit covers the following: • Need(s) for Engineering Services, Goods and Works. • Tender Processes. • Contract Awards.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Need(s) for Engineering Services, Goods, and Works	To be competent, the individual must be able to: PC1. Identify the need(s) for engineering services (consultancy), goods, and works. PC2. Identify the most appropriate method to employ for procurement of services, goods or works. PC3. Engage the key stakeholders for input on the need(s). PC4. Formulate the appropriate procurement plans. PC5. Identify the most appropriate methods of procurement.
Tender Processes	To be competent, the individual must be able to: PC6. Design solicitation documents (contract, bills of quantities, specifications, drawings, other schedules; technical reports, etc.) that respond to the identified need(s). PC7. Ensure that the procurement is done in line with ZPPA guidelines. PC8. Establish confidential cost estimates for budgetary purposes. PC9. Promulgation of the tender process. PC10. Adjudicate tender. PC11. Produce tender evaluation report. PC12. Participate in tender negotiations. PC13. Prepare tender negotiation minutes and addenda. PC14. Compile and file contract documents.
Contract Awards	To be competent, the individual must be able to: PC15. Generate letter of award to the successful bidder. PC16. Facilitate receipt of acceptance letter from the Bidder to the client/owner. PC17. Assist in the award of the contract(s).
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant standards, procedures, and policies of the organisation relating to procurement.

	OK2. Context of the organisation as determined by whether it is client/owner's body, consultancy or construction, as well as legal, financial, social, regulatory, and cultural. OK3. Internal factors, organisation structure, governance, and resource capabilities. OK4. Roles and responsibilities. OK5. Risk management. OK6. Occupational safety, health, quality, and environmental management. OK7. Performance evaluation. OK8. Change order management.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of, but not limited to, the following: TK1. Technical knowledge regarding the identified need(s). TK2. Relevant design codes and standards. TK3. Ability to read technical drawings. TK4. Computer literacy. TK5. Comprehension of civil engineering standard form contract types. TK6. Knowledge of procurement methods, such as Design-Bid-Build, Design and Build, PPP, etc. TK7. Knowledge of drafting contract documents and associated addenda. TK8. Comprehension of relevant technical specifications for materials, performance, and conformity. TK9. Understand the technical Knowledge of various construction materials. TK10. Apply basic risk management. TK11. Apply basic green design principles. TK12. Employ basic negotiations strategies. TK13. Knowledge of required staffing levels. TK14. Collaborate with other specialisations, such as mechanical, electrical/electronics, geomatic engineering, Quantity Surveyors, Town and Country Planners, Architects, Archaeologists, etc. TK15. Demonstrate knowledge and ability to respond to queries requiring clarifications of the tender documentation.
C. Regulatory Context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of regulatory compliance aspects of the industry pertaining to latest laws, regulations, guidelines and specifications, but not limited to: RK1. Engineering Institution of Zambia Act, 2010. RK2. Mines and Minerals Development Act, 2015. RK3. Environmental Management Act, 2011. RK4. Occupational Health and Safety Act, 2010. RK5. Factories Act Cap 441. RK6. Workers Compensation Act No. 10 of 1999. RK7. Employment Act Chapter 268, and associated labour legislature. RK8. Public Roads Acts, No. 12 of 2012.

	RK9. Tolls Act 2011. RK10. Public Finance Management Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Produce evaluation report(s). CS2. Prepare and provide clear and simple instructions, details, and sketches to co-workers/others. CS3. Develop/review standard particular specifications. CS4. Develop/review particular contract conditions. CS5. Communicate effectively through writing as appropriate for the benefit of the audience. CS6. Use conversational communication methods such as e-mail, WhatsApp, etc CS7. Demonstrate good command of the English language
	Reading Skills
	The individual on the job must be able to: CS8. Read English, and/or be able to give clear instructions. CS9. Read and cross-check that tender documentation is responsive to the requirements. CS10. Research, read, and interpret technical data from manuals, books, and other relevant literature. CS11. Read and comprehend written information or communication.
B. Professional Skills	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS12. Demonstrate active listening and interpret communication/instructions from the supervisor, co-workers and tenderers. CS13. Convey information clearly and concisely to co-workers and others. CS14. Illustrate ideas using presentation skills and applications such as PowerPoint, etc. CS15. Manage pre-tender meetings and discussions. CS16. Manage tender evaluation and negotiations meetings. CS17. Provide feedback on technical works and reports.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Adhere to the organisation's decision-making policies. PS2. Make independent and sound decisions based on engineering judgement, and considering public safety and interest of other stakeholders. PS3. Make prompt decisions on feedback on tender documentation evaluations. PS4. Demonstrate impartiality in deciding the responsive tenderer. PS5. Demonstrate high ethical standards.

	Plan and Organise
	The individual on the job should be able to: PS6. Plan, organise, lead, and control the organisation's activities in line with the applicable deadlines. PS7. Schedule tasks and complete them within the allocated time and budget. PS8. Work to achieve set goals. PS9. Evaluate the tender documentation in line with the tender procurement plan.
	Customer Centricity
	The individual on the job should be able to: PS10. Manage customer relations with intent to satisfying their project requirements.
	Problem-Solving
	The individual on the job should be able to: PS11. Demonstrate ability to identify problems, brainstorm, and analyse possible results with the view of implementing the optimal solutions. PS12. Consult widely and identify possible remedies. PS13. Escalate when required in line with the organisation's procedures and protocols.
	Analytical Thinking
	The individual on the job should be able to: PS14. Apply domain knowledge, observations, and data to perform tasks related to the assignment. PS15. Apply methodical step-by-step approach to thinking, and breaking down complex problems into smaller and manageable components. PS16. Demonstrate creativity, interpersonal, and organisational skills. PS17. Employ data analysis, logical thinking, research, and communication skills.
	Critical Thinking
	The individual on the job should be able to: PS18. Observe and predict opportunities, threats, and solutions. PS19. Collect, understand, interpret data and other information. PS20. Demonstrate ability to draw inferences based on relevant data, personal knowledge, and experience. PS21. Communicate resulting information with others verbally, non-verbally, and in writing.

UNIT 3 [This Unit covers the skills and knowledge required by a Civil Engineer to undertake execution of civil engineering works]

Unit No.	03
Unit Title	Execution of Civil Engineering Works
Description	This Unit describes the skills and knowledge required by a Civil Engineer for the execution of civil engineering works.
Scope	This Unit covers the following: • Project Implementation. • Project Management. • Construction Management. • Contract Management. • Testing and Commissioning. • Occupational Safety Health and Environment Management.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Project Implementation	To be competent, the individual must be able to: PC1. Identify the need(s) for engineering services (consultancy), goods, and works. PC2. Comprehend and identify the most appropriate methods to employ for procurement of services, works or goods. PC3. Engage the key stakeholders for input on the need(s). PC4. Formulate the appropriate procurement plan. PC5. Identify the most appropriate methods of procurement.
Project Management	To be competent, the individual must be able to: PC6. Demonstrate knowledge of the project cycle. PC7. Appreciate the work breakdown structure. PC8. Employ cost estimation and budget tracking. PC9. Apply vendor management. PC10. Perform project scheduling (programming). PC11. Carry out quality control. PC12. Employ project controls for time, cost, scope, and risk. PC13. Undertake human resource management. PC14. Perform project communication management. PC15. Execute project records management. PC16. Oversee project design, and construction.
Construction Management	To be competent, the individual must be able to: PC17. Demonstrate knowledge of overall planning, coordination and control of the construction process from start to finish. PC18. Organise and manage project budget. PC19. Set and track the schedules and ensure all tasks are completed on time. PC20. Oversee site safety and security. PC21. Ensure that the project is completed on time, on budget, and to client specifications. PC22. Conduct change order management. PC23. Manage construction resources (human capital, plant and equipment, tools, materials, etc.)

	PC24. Perform quality control management PC25. Draw up an operation and maintenance plan. PC26. Manage key stakeholders. PC27. Coordinate the multidisciplinary team on the project. PC28. Facilitate safety, health, and environmental management PC29. Conduct and manage site progress and technical meetings. PC30. Oversee communication management. PC31. Implement project records management. PC32. Undertake project close out.
Contract Management	To be competent, the individual must be able to: PC33. Comprehend Contract Law. PC34. Demonstrate basic comprehension of a Construction Contract. PC35. Formulate a Construction Contract document. PC36. Demonstrate understanding of various standards and model forms of contracts. PC37. Manage contract execution. PC38. Carry out risk identification, allocation, and management. PC39. Manage contract claims. PC40. Undertake change order management. PC41. Comprehend the roles, relationships, and responsibilities of the parties to the contract. PC42. Implement project records management. PC43. Demonstrate understanding of breach of contract. PC44. Appreciate various types of contract securities (bonds, guarantees, and insurance). PC45. Facilitate dispute and conflict management.
Testing and Commissioning	To be competent, the individual must be able to: PC46. Plan the testing and commissioning procedure. PC47. Conduct all requisite testing on the completed construction as specified to verify conformance to design and specifications. PC48. Perform the on-site commission. PC49. Prepare the testing and commissioning report.
Occupational Safety Health and Environmental Management (OSHE)	To be competent, the individual must be able to: PC50. Demonstrate understanding of Occupational Safety, Health, and Environmental (OSHE) aspects. PC51. Appreciate the company safety, health, and environmental policies and procedures. PC52. Conduct the job hazard safety analysis. PC53. Comprehend fire hazards. PC54. Facilitate on-site environmental management. PC55. Compile a safety management plan. PC56. Guide the installation of appropriate safety signage on site. PC57. Appreciate the importance of Toolbox Talk and other safety training. PC58. Enforce usage of PPE.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: - OK1. Project, construction, and contract management. - OK2. Relevant standards, procedures, and policies of the organisation relating to OSHE. - OK3. Consultancy or construction, as well as legal, financial, social, regulatory and cultural context of the organisation. - OK4. Internal factors, organisation structure, governance, and resource capabilities. - OK5. Roles and responsibilities. - OK6. Risk management. - OK7. OSHE management. - OK8. Performance evaluation. - OK9. Change order management.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding with ability to: - TK1. Comprehend project, construction, and contract management - TK2. Demonstrate knowledge regarding the OSHE aspects, and the relevant codes and standards. - TK3. Demonstrate computer literacy skills. - TK4. Formulate contract documentation and associated OSHE components. - TK5. Compile relevant technical specifications for materials, performance, conformity, and impact on OSHE. - TK6. Interpret Technical Knowledge of various construction materials. - TK7. Employ basic risk management knowledge. - TK8. Apply basic green design principles. - TK9. Assemble the required staffing levels for a particular task - TK10. Collaborate with other specialisations, such as mechanical, electrical/electronics, Quantity Surveyors, Architects, etc. - TK11. Demonstrate ability to facilitate drafting of the OSHE management plan. - TK12. Conduct quality control and testing.
C. Regulatory Context (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of regulatory compliance aspects of the industry pertaining to latest laws, regulations, guidelines, and specifications, but not limited to those listed below: - RK1. Engineering Institution of Zambia Act, 2010. - RK2. Mines and Minerals Development Act, 2015. - RK3. Environmental Management Act, 2011. - RK4. Occupational Health and Safety Act, 2010. - RK5. Factories Act Cap 441. - RK6. Workers Compensation Act No. 10 of 1999. - RK7. Employment Code Act. - RK8. Public Roads Acts, No. 12 of 2012.

	RK9. Tolls Act 2011. RK10. Public Procurement Act 2020, etc.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills The individual on the job must be able to: CS1. Produce progress, technical, claims, safety, environmental, final completion, maintenance, and operations reports, etc. CS2. Prepare and provide clear instructions, details, and sketches to co-workers/others. CS3. Review standard particular specifications relating to OSHE. CS4. Review particular contract conditions relating to OSHE. CS5. Communicate effectively through writing as appropriate for the benefit of the audience. CS6. Use conversational communication methods, such as e-mail, WhatsApp, etc. CS7. Demonstrate good command of the English language. CS8. Facilitate Toolbox Talk write-up and other safety and environmental training materials.
	Reading Skills The individual on the job must be able to: CS9. Read English, and be able to give clear instructions. CS10. Research, read, and interpret technical data from manuals, books, and other relevant literature. CS11. Read and comprehend written information or communication.
	Oral Communication (Listening and Speaking Skills) The individual on the job must be able to: CS12. Demonstrate active listening and interpret communication/instructions from the supervisor, co-workers and tenderers. CS13. Convey information clearly and concisely to co-workers and others. CS14. Illustrate ideas using presentation skills and applications such as PowerPoint, etc. CS15. Manage pre-tender meetings and discussions. CS16. Manage tender evaluation and negotiations meetings. CS17. Provide feedback on technical works and reports.
	Decision-Making The individual on the job must be able to: PS1. Adhere to the organisation's decision-making policies. PS2. Make independent and sound decisions based on engineering judgement, and considering public safety and interest of other stakeholders. PS3. Make prompt decisions on feedback of tender documentation evaluations.
B. Professional Skills	

	PS4. Demonstrate impartiality in deciding the responsive tenderer. PS5. Demonstrate high ethical standards.
	Plan and Organise
	The individual on the job should be able to: PS6. Plan, organise, lead, and control the organisation's activities in line with the applicable deadlines. PS7. Schedule tasks and complete them within the allocated time and budget. PS8. Work to achieve set goals.
	Customer Centricity
	The individual on the job should be able to: PS9. Manage customer relationships with intent to satisfying their project requirements.
	Problem-Solving
	The individual on the job should be able to: PS10. Demonstrate ability to identify problems, brainstorm, and analyse possible results with the view of implementing the optimal solutions. PS11. Consult widely and identify possible remedies. PS12. Escalate when required in line with the organisation's procedures and protocols.
	Analytical Thinking
	The individual on the job should be able to: PS13. Apply domain knowledge, observations, and data to perform tasks related to the assignment. PS14. Apply a methodical step-by-step approach to thinking and breaking down complex problems into smaller and manageable components. PS15. Demonstrate creativity, interpersonal skills, and organisational skills. PS16. Employ data analysis, logical thinking, and research and communication skills.
	Critical Thinking
	The individual on the job should be able to: PS17. Observe and predict opportunities, threats and solutions. PS18. Collect, understand, interpret data and other information. PS19. Demonstrate ability to draw inferences based on relevant data, and personal knowledge and experience. PS20. Communicate resulting information with others verbally, non-verbally, and in writing.

UNIT 4 [This Unit covers the skills and knowledge required by a Civil Engineer to undertake operations and maintenance of civil engineering works]

Unit No.	04
Unit Title	Operations And Maintenance
Description	This Unit describes the skills and knowledge required by Civil Engineer to execute operations and maintenance of civil engineering works.
Scope	This Unit covers the following: • Operations • Maintenance
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Operations	To be competent, the individual must be able to: PC1. Formulate as-built drawings/information, and operations/user Manuals for the facility, infrastructure, etc. PC2. Conduct training for staff responsible for operations. PC3. Operate the facility, infrastructure, etc.
Maintenance	To be competent, the individual must be able to: PC4. Generate maintenance manuals. PC5. Carry out repairs and maintenance on the facility or infrastructure, etc.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant standards, procedures, and policies of the organisation relating to operations and maintenance. OK2. Consultancy or construction, as well as legal, financial, social, regulatory and cultural context of the organisation. OK3. Internal factors, organisational structure, governance, and resource capabilities. OK4. Roles and responsibilities. OK5. Occupational Health and Safety, Quality and Environmental management.
B. Technical Knowledge	The individual on the job must be able to: TK1. Demonstrate technical knowledge regarding operations and maintenance. TK2. Apply the relevant design codes and standards. TK3. Read technical drawings. TK4. Demonstrate computer literacy. TK5. Demonstrate understanding relevant technical specifications for materials, performance, and conformity. TK6. Comprehend technical knowledge of various construction materials. TK7. Apply basic risk management. TK8. Apply basic green design principles. TK9. Undertake human resource management.

C. Regulatory Context (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of regulatory compliance aspects of the industry pertaining to latest laws, regulations, guidelines, and specifications, but not limited to: RK1. Engineering Institution of Zambia Act, 2010. RK2. Mines and Minerals Development Act, 2015. RK3. Environmental Management Act, 2011. RK4. Occupational Health and Safety Act, 2010. RK5. Factories Act Cap 441. RK6. Workers Compensation Act No. 10 of 1999. RK7. Employment Code Act. RK8. Public Roads Acts, No. 12 of 2012. RK9. Tolls Act 2011. RK10. Public Procurement Act, 2020, etc.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Produce operations and maintenance manuals, technical reports, etc. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers/others. CS3. Communicate effectively through writing as appropriate for the benefit of the audience. CS4. Use conversational communication methods such as e-mail, WhatsApp, etc. CS5. Demonstrate good command of the English language.
	Reading Skills
	The individual on the job must be able to: CS6. Read English, and be able to give clear instructions. CS7. Research, read, and interpret technical data from manuals, books, and other relevant literature. CS8. Read and comprehend written information or communication.
B. Professional Skills	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS9. Demonstrate active listening and interpret communication/instructions from the supervisor, co-workers. CS10. Convey information clearly and concisely to co-workers and others. CS11. Illustrate ideas using presentation skills and applications such as PowerPoint, etc. CS12. Provide feedback on technical works and reports.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Adhere to the organisation's decision-making policies.

	PS2. Make independent and sound decisions based on engineering judgement, and considering public safety and interest of other stakeholders. PS3. Make prompt decision on feedback related to operations and maintenance. PS4. Demonstrate high ethical standards.
	Plan and Organise
	The individual on the job should be able to: PS5. Plan, organise, lead, and control organisation activities in line with the applicable deadlines. PS6. Schedule tasks and complete them within allocated time and budget. PS7. Work to achieve set goals.
	Customer Centricity
	The individual on the job should be able to: PS8. Manage customer relationships with intent to satisfying their project requirements.
	Problem-Solving
	The individual on the job should be able to: PS9. Demonstrate ability to identify problems, brainstorm, and analyse possible results with the view of implementing the optimal solutions. PS10. Consult widely and identify possible remedies. PS11. Escalate when required in line with the organisation's procedures and protocols.
	Analytical Thinking
	The individual on the job should be able to: PS12. Apply domain knowledge, observations, and data to perform tasks related to the assignment. PS13. Apply a methodical step-by-step approach to thinking and breaking down complex problems into smaller and manageable components. PS14. Demonstrate creativity, interpersonal, and organisational skills. PS15. Employ data analysis, logical thinking, and research and communication skills.
	Critical Thinking
	The individual on the job should be able to: PS16. Demonstrate ability to observe and predict opportunities, threats, and solutions. PS17. Collect, understand, interpret data and other information. PS18. Demonstrate ability to draw inferences based on relevant data, and personal knowledge and experience. PS19. Communicate resulting information with others verbally, non-verbally, and in writing.

UNIT 5 [This Unit covers the skills and knowledge required by a Civil Engineer to perform leadership roles and apply supervisory skills]

Unit No.	05
Unit Title	Application of Leadership Roles and Supervisory Skills
Description	This Unit describes the skills and knowledge required by a Civil Engineer in demonstrating leadership roles and supervisory skills of civil engineering works.
Scope	This Unit covers the following: • Delegation of Duties and Responsibilities; Supervision. • Training, Mentorship, and Team Building.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Delegation of Duties and Responsibilities; Supervision	To be competent, the individual must be able to: PC1. Demonstrate ability to explain clearly the tasks to be delegated to the subordinates. PC2. Delegate duties and responsibilities to the appropriate subordinates, at the right time, with the necessary resources, skills, and authority to complete supervisory tasks effectively. PC3. Collaborate in setting clear and achievable performance criteria, monitor progress, and provide feedback.
Training, Mentorship, and Team Building	To be competent, the individual must be able to: PC4. Identify the skills gap, and provide appropriate training and mentorship to the subordinates. PC5. Conduct team building to promote team spirit.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant standards, procedures, and policies of the organisation. OK2. Organisational roles, responsibilities, accountabilities, and authorities. OK3. Organisational strategic goals and objectives.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Planning, organising, directing, and controlling. TK2. Problem-solving and decision-making. TK3. Conflict resolution management. TK4. Effective communication. TK5. Knowledge of required staffing levels. TK6. Performance management.
C. Regulatory Context (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of regulatory compliance aspects of the industry pertaining to latest laws, regulations, guidelines, and specifications but not limited to: RK1. Engineering Institution of Zambia Act, 2010. RK2. Mines and Minerals Development Act, 2015.

	RK3. Environmental Management Act, 2011. RK4. Occupational Health and Safety Act, 2010. RK5. Factories Act Cap 441. RK6. Workers Compensation Act No. 10 of 1999. RK7. Add Employment Code Act. RK8. Public Roads Acts, No. 12 of 2012. RK9. Tolls Act 2011. RK10. Public Procurement Act 2020, etc.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Produce training and mentorship manuals, performance evaluation reports, etc. CS2. Prepare and provide clear and simple instructions, details, and sketches to co-workers/others. CS3. Communicate effectively through writing as appropriate for the benefit of the audience. CS4. Use conversational communication methods such as e-mail, WhatsApp, etc. CS5. Demonstrate good command of the English language.
	Reading Skills
	The individual on the job must be able to: CS6. Read English, and be able to give clear instructions. CS7. Research, read, and interpret technical data from manuals, books, and other relevant literature. CS8. Read and comprehend written information or communication.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS9. Demonstrate active listening and interpret communication/ instructions from the supervisor, co-workers. CS10. Convey information clearly and concisely to co-workers and others. CS11. Illustrate ideas using presentation skills and applications such as PowerPoint, etc. CS12. Provide feedback on delegated tasks.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Adhere to the organisation's decision-making policies. PS2. Make independent and sound decisions based on engineering judgement, and considering public safety, subordinates, and interest of other stakeholders. PS3. Make prompt decisions on feedback on delegated tasks. PS4. Demonstrate high ethical standards.

	Plan and Organise
	The individual on the job should be able to: PS5. Plan, organise, lead, and control the organisation's activities in line with the applicable deadlines. PS6. Schedule training tasks. PS7. Work to achieve set goals. PS8. Employ SWOT analysis to evaluate the performance of the delegated tasks, training, mentorship, and team building activities.
	Customer Centricity
	The individual on the job should be able to: PS9. Provide leadership and supervision roles with the aim to enhance customer relationships with intent to satisfying their project requirements.
	Problem-Solving
	The individual on the job should be able to: PS10. Demonstrate ability to identify problems, brainstorm, and analyse possible results with the view of implementing the optimal solutions. PS11. Consult widely and identify possible remedies. PS12. Escalate when required in line with the organisation's procedures and protocols.
	Analytical Thinking
	The individual on the job should be able to: PS13. Apply domain knowledge, observations, and data to perform tasks related to the assignment. PS14. Apply a methodical step-by-step approach to thinking and breaking down complex problems into smaller and manageable components. PS15. Demonstrate creativity, interpersonal, and organisational skills. PS16. Employ data analysis, logical thinking, research, and communication skills.
	Critical Thinking
	The individual on the job should be able to: PS17. Demonstrate ability to observe and predict opportunities, threats, and solutions. PS18. Collect, understand, interpret data and other information. PS19. Demonstrate ability to draw inferences based on relevant data, and personal knowledge and experience. PS20. Communicate resulting information with others verbally, non-verbally, and in writing.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to:

Equipment and Tools: The individual on the job should be familiar with, and be able to operate the following equipment or their latest equivalent:

Survey Works: Automatic level, Laser level, Total station, Laser range Finder, Hand tools (handheld GPS, measuring tape, spirit level, square etc.)

Field/Laboratory: Basic field and laboratory testing equipment

Software: Structural Work: Basic knowledge of structural design packages

Civil Work: Basic knowledge of Civil and Digital Terrain Modelling software.

Office Works: MS Suite (Word, Excel, PowerPoint, MS Project, Access, etc.)

Raw Materials and Consumables: Basic knowledge of modern construction materials.

6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Civil Engineers face challenges such as: obsolete and/or inappropriate equipment and tools; budgetary constraints; inadequate product costing skills; poor technical skills base; bureaucracy in procurement procedures; lack of appreciation of preventive maintenance by non-engineering management staff; labour intensive nature of the work; rapid change of technology and materials; lack of personal protective equipment; climate change; cyber warfare; inconsistencies in company and government policies and regulations, etc. In some cases, a Civil Engineer encounters equipment that is more advanced than they were trained on.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to challenges include: selecting and procuring appropriate equipment and tools for the job; supporting capacity building through training; identifying and utilising suitable adaptation and mitigation measures against the effects of climate change; utilising appropriate cyber security measures to protect against cyber warfare; inclusion of engineering professionals in management teams; deployment of automation where feasible; provision of PPE; participating in lobbying and formulation of policies; allocation of adequate financial resources, etc.

7. WORKING CONDITIONS/ENVIRONMENT

Civil Engineers work with a variety of construction equipment, toxic substances, and volatile materials. Their work environment is susceptible to fires, explosions, structural failures, and equipment malfunctions. Working conditions include: cold, hot, and wet conditions; climbing great heights; standing/walking for long hours; lifting materials; working in day or night shifts; areas that are noisy and dusty; areas with limited lighting or ventilation, etc.

8. PARTIES INVOLVED/INTERACTING WITH THE JOBHOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Management, supervisors, subordinates, and other section members, etc.

8.2 External/Outside the Organisation

Government regulators, professional bodies, clients, suppliers, engineers from other companies, labour unions, students/interns, etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Be able to sit for sustained period of time at the design desk;
- Bend, stretch, twist, or reach out;
- Be able to lift relatively heavy materials, tools, and equipment;
- Be able to use fingers, hands, and feet with ease to complete the assigned task (dexterity); etc.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part, and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.CE.01		
ZQF Level	7	Version Number	01
Sector	Construction	Date of Approval	December 2023
Sub Sector	Building Works, Roads and Highways, Bridges, Railways, Airports, Ports, Renewable Energy, Infrastructure, Water and Sanitation, Dams and Tunnels, and Irrigation.	Date of Last Review	N/A
Occupation	Civil Engineering	Date of Next Review	December 2028



*“You learn,
We Standardise”*

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